

Controlled chemical worlds reveal limits of generalization in autonomous scientific research

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Abstract

Autonomous scientific research requires deciding which experimental relationships generalize to new conditions. This ability is difficult to evaluate when mechanisms are unknown and agents select their own evidence. Here we introduce ChemWorld, a programmable framework separating chemical process laws, experimental observations and supplied information. Across 240 autonomous campaigns in six system families, operational achievement and predictive knowledge diverge. Electrochemical optimization improves recommendation performance in 26 of 30 matched comparisons, but prediction in only 14. Larger budgets improve electrochemical and partitioning predictions, while crystallization errors remain response dependent. In aqueous equilibrium, aligned information improves predictions on nine questions but worsens them on the three most dilute conditions in every tested world. Conversely, crystallization predictions unnecessarily depart from stable observed purity. These complementary failures identify the applicability of empirical relationships as a central evaluation problem for autonomous science, beyond successful operation or plausible mechanistic explanation. Autonomous laboratories and language-model agents increasingly connect scientific planning to executable experiments¹⁻³. Their scientific promise is to turn those experiments into relationships that remain useful under new interventions. Finding a productive condition is one outcome of this process. Predicting what will happen elsewhere also requires judging which observed relationships should persist, which may change, and how much uncertainty remains. The applicability of what a researcher learns is therefore a distinct object of evaluation.

This capability is difficult to evaluate in physical laboratories. Mechanisms are incompletely known, starting states are hard to reproduce, and researchers select their own evidence. Prior information can change both the experiments performed and their interpretation. A failed prediction may reflect missing discriminating experiments, unresolved identifiability or an extrapolation poorly supported by the observations. Conversely, a correct prediction may exploit a simple regularity without identifying its mechanism. Task scores and the apparent sophistication of a scientific report do not resolve these alternatives.

ChemWorld provides a computational instrument for studying this complete research process. Optimization benchmarks offer repeatable objectives^{4,5}, while interactive chemistry and discovery environments support sequences of scientific actions⁶⁻⁸. We combine composable process laws, persistent experimental state and independently controlled public information with autonomous experiment selection. The evaluator can trace actual operations and observations and assess the original researcher’s subsequent predictions, preserving the connection between acquiring evidence and using it.

Across six chemical system families, we first show that improved operation and predictive generalization can separate. Larger research envelopes improve several outcomes, and correct partial information has substantial benefits in some conditions. Those benefits nevertheless depend on the response and intervention regime. Equilibrium and crystallization then expose complementary patterns: some forecasts extend a local plateau into a regime where it fails, while others depart from an observed response that remains stable. Together, these results motivate evaluating how autonomous researchers assign an applicability domain to the relationships formed during their experiments.

Results

A programmable experimental instrument separates laws, evidence and information

ChemWorld constructs executable worlds from compatible process and observation components. Each world specifies persistent material states, legal operations, instruments, finite resources and private parameters. Operations can change composition, phase allocation, thermal history and remaining material; measurements expose declared quantities rather than the full simulator state. The evaluator can vary private laws independently of public descriptions, or change supplied information while keeping a world fixed (Fig. 1).

The research programme contains 240 campaigns, each belonging to a five-world, three-arm study block. Opaque supplies usable public contracts and anonymous identifiers. Aligned adds the tested nominal information; MisIndexed supplies its fixed mismatch or reassignment. Information can concern entity associations, effective parameters or structural relationships. Aligned information is partial, and Opaque does not remove general chemical knowledge. These distinctions avoid equating any dossier with complete knowledge of the simulator.

Each campaign follows the original agent from experiment selection to scientific assessment. Within a finite research envelope, it chooses conditions, repetitions and measurements and receives their physical consequences. Where a task requires an operating recommendation, that recommendation is sealed before the free-form mechanism report (K1) and evaluated by an independent retest. The same agent then predicts withheld conditions and uncertainty intervals (Q), retaining its full research context. Retrospective questions (K2) follow the sealed predictions without revealing the reference outcomes. This chronology separates what was stated in the report from what becomes explicit during prediction.

The agent freely allocates 12 or 24 batches and a matching allowance of additional measurements. Objectives are tailored to the system: discovery and optimization are separate commissions in electrochemistry and reaction processing; partitioning and equilibrium emphasize characterization; crystallization and purification seek recovery subject to product-quality constraints. All 720 report, prediction and retrospective stages completed. Two campaigns fell short of their final-assay budgets, leaving 238 conforming chains and 3,597 of 3,600 planned source assays. Both remain in the results.

Finite-domain platform qualification establishes execution and accounting consistency, with detailed coverage reported in Methods. Exact environment replay reconstructs the actions and observations; it does not reconstruct unrecorded internal reasoning or establish wet-laboratory validity. This design makes operational achievement, predictive generalization and uncertainty separately measurable within the same autonomous research process.

A selected crystallization pair makes this research process concrete (Fig. 2). The independent 12-batch session finds a quality-feasible product in batch 1 and refines it by lowering the cooling endpoint; the 24-batch session first becomes feasible in batch 20 after changing heating and introducing staged cooling. Their selected procedures are independently retested. Recorded operations, observed outcomes and later public explanations remain distinct: the logs do not provide contemporaneous decision reasons. This selected pair illustrates the process, not a population budget effect.

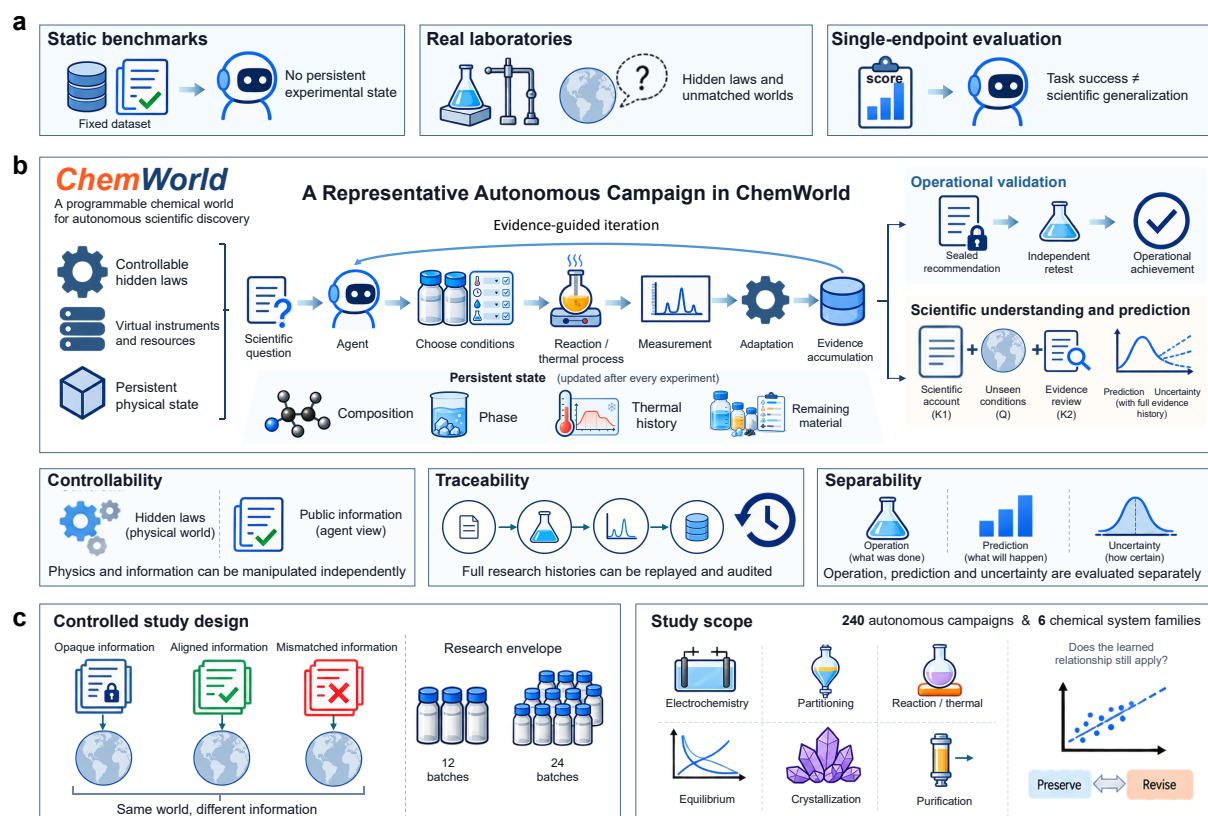


Figure 1: ChemWorld as a controlled instrument for autonomous scientific research. **a**, Evaluation challenges posed by fixed datasets, physical laboratories and single-endpoint task scores. **b**, An autonomous campaign connects experimental choice, process execution, measurement and evidence-guided adaptation in a persistent physical world. Hidden laws and public information can be varied independently, research histories can be replayed, and operation, prediction and uncertainty are assessed separately. Where required, a recommendation is sealed and independently retested. The original session then provides a scientific account (K1), sealed predictions (Q) and retrospective evidence review (K2), in that order with its research history available. Neither Q nor K2 receives target feedback. **c**, Matched-world information conditions, independent 12- and 24-batch research envelopes, and the completed study's 240 campaigns across six system families. "Mismatched information" denotes MisIndexed. Budget comparisons apply to the systems with both research envelopes.

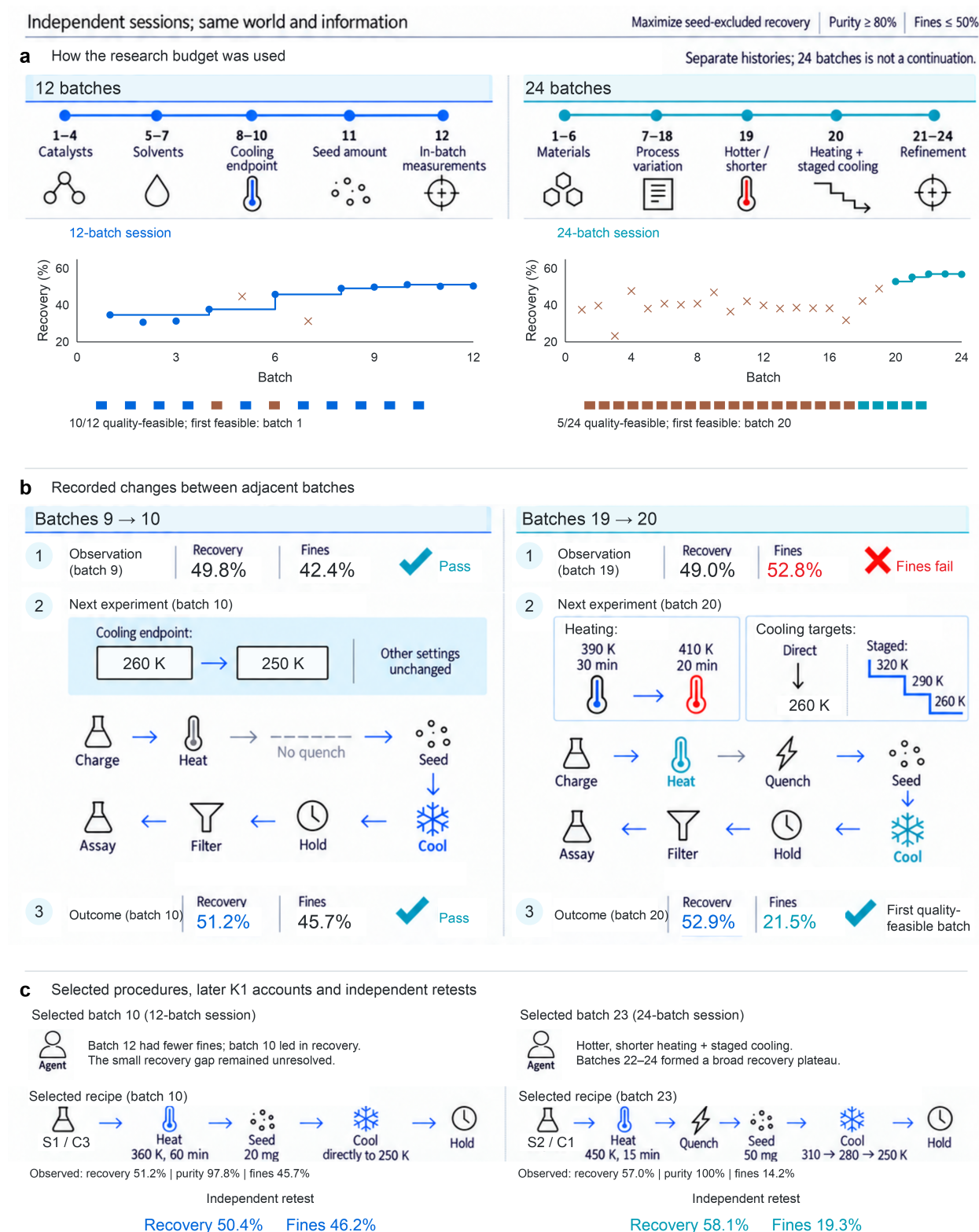


Figure 2: Two original research paths in one matched crystallization world. a, Independent 12- and 24-batch sessions. Circles and crosses retain feasible and infeasible observations; staircases show best feasible recovery, undefined before first feasibility. Strips mark each batch's quality status. b, Recorded changes between adjacent batches; coloured operations indicate revised settings. Quench stops reaction chemistry while cooling and crystal growth remain possible. c, Selected procedures, condensed later K1 accounts, source observations and independent retests. Quality requires purity at least 0.80, fines at most 0.50 and particles present. Recovery excludes seed mass; fines are particles below 20 micrometres. Temperatures denote requested targets. Immediate thoughts were not recorded. This selected case does not establish a population budget effect.

Operational achievement does not imply predictive generalization

We first ask whether a research commission that produces a better operating recommendation also produces better predictions. Discovery and optimization campaigns are paired within a world, information arm, budget and prior locus, so the comparison follows the outcomes of two complete autonomous research processes.

In electrochemistry, the optimization commission raises the mean independently retested score from 0.543 to 0.706, with higher world-level means in all five worlds (Fig. 3a). Mean score-prediction error is nevertheless almost unchanged, from 0.1430 to 0.1431 (Fig. 3b). This aggregate contrast does not imply equivalence of the predictions. At the matched-pair level, recommendation performance improves in 26 of 30 comparisons, whereas score-prediction error improves in 14. Thirteen pairs improve on both endpoints; another thirteen combine a better recommendation with worse prediction (Fig. 3c). These discordances occur in all five worlds, in one to four of six pairs per world (Fig. 3d). Restricting to pairs without infrastructure-driven source restarts retains the separation: 15 of 17 improve recommendation performance, eight improve prediction and seven are discordant.

The effect is not universal. In reaction processing, optimization improves retests in nine of 30 comparisons, score prediction in eight and the original six-response macro error in six. Seventeen pairs become worse on both retest and score prediction (Fig. 3c). Thus the optimization commission itself does not guarantee better delivery. Prediction and decision losses need not align⁹, and these experiments show why both must be measured in an autonomous research workflow.

Operational quality constraints provide a further distinction between recovery and acceptable delivery in purification (Supplementary Fig. S5a).

These results establish the need to measure operational and predictive outcomes separately. They leave open which aspects of generalization can improve when the researcher receives more resources.



Figure 3: Operational gains do not guarantee predictive gains. a,b, Electrochemical recommendation retest scores (higher is better) and score-prediction mean absolute error (MAE; lower is better). Each world-level bar averages six campaigns per commission: three information arms and two budgets at the entity prior locus. Header means summarize all thirty campaigns per commission. c, All four joint outcomes of optimization relative to discovery, counted across thirty matched pairs per system. Reaction-processing pairs span five worlds, three arms and two prior loci at twelve batches. Pairing fixes world, arm, budget and prior locus; better or worse denotes any strictly positive or negative change, without a significance threshold. There are no exact ties. d, Electrochemical pairs with better retests but worse prediction, out of six per world. Panels a,b show means; c,d retain individual pair outcomes. All 120 campaigns are retained; pairs share worlds and are not independent world replicates. The comparison does not isolate experiment selection from subsequent inference.

Larger research envelopes improve generalization selectively

We next compare 12- and 24-batch research envelopes under matched worlds, information arms and commissions. The larger condition is a new autonomous session with increased experimental and computational allowances, rather than a continuation of the smaller campaign.

Increasing the research envelope from 12 to 24 batches substantially improves several predictive outcomes (Fig. 4). Electrochemical score mean absolute error (MAE) decreases from 0.17378 to 0.11222 under discovery and from 0.17811 to 0.10818 under optimization. Eleven and twelve of the respective fifteen matched campaign pairs improve. All six electrochemical responses improve in aggregate, and their mean directions persist when each world is omitted in turn.

Phase-partitioning organic-fraction MAE decreases from 0.08291 to 0.02379, with improvement in 13 of 15 pairs. Correct process-choice answers increase from 25 of 30 to 30 of 30. Nominal 90% interval coverage rises from 78.3% to 98.9% while mean width narrows from 0.25122 to 0.18553. This combination supports better prediction, although coverage above the nominal level is not perfect calibration.

Crystallization behaves differently. Net-recovery MAE decreases from 0.10433 to 0.07937, but quality-eligible recommendations remain 12 of 15 at each budget and mean retested recovery decreases from 0.42054 to 0.39939. Fines-fraction interval coverage improves from 35.6% to 43.9%, still below the nominal 80% in every campaign.

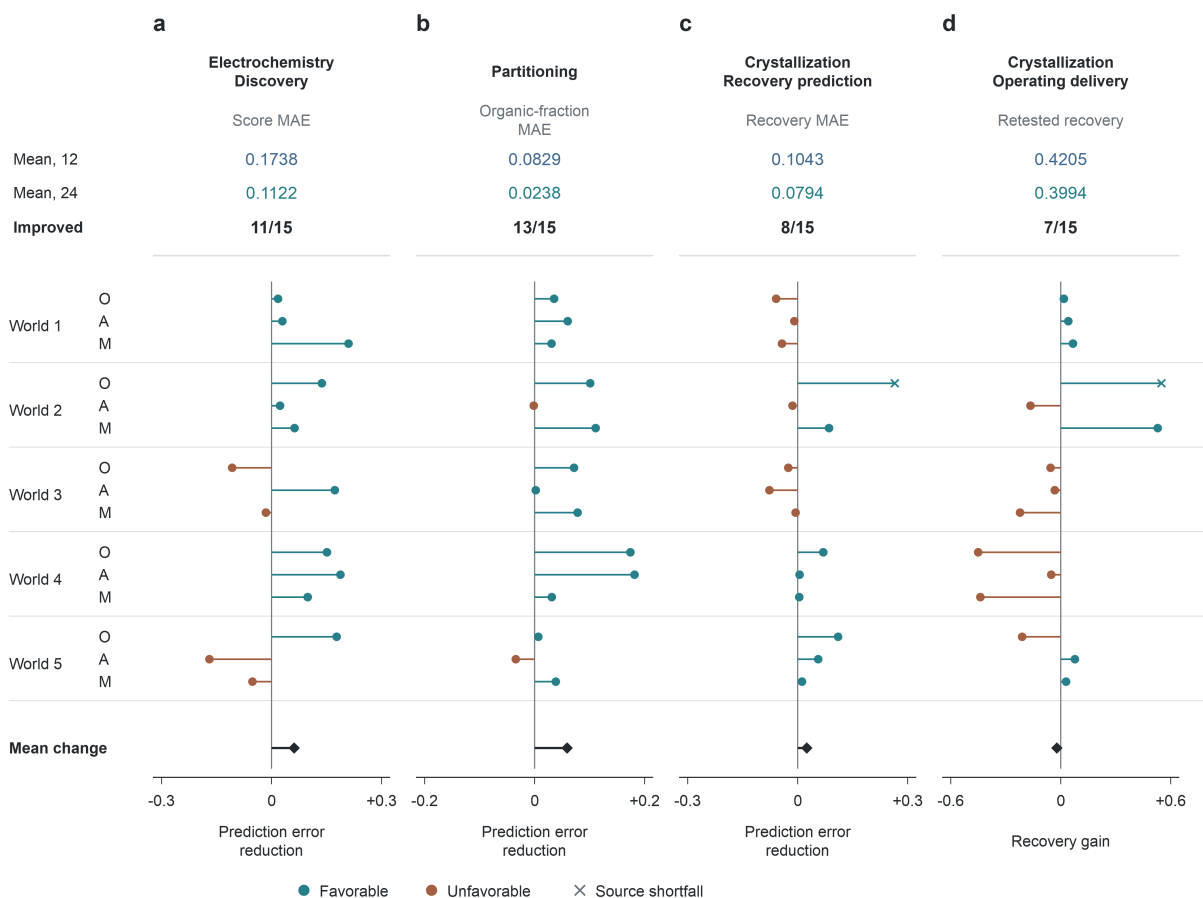


Figure 4: Larger research envelopes have response-specific effects. Headers report campaign means at 12 and 24 batches and counts of improved pairs. Each row retains one world-arm pair. Panels a-c show prediction MAE at 12 minus MAE at 24; panel d shows retested recovery at 24 minus recovery at 12. Positive change therefore favours the larger envelope. Diamonds show mean changes; crosses retain pairs with a source-assay shortfall. O, A and M denote Opaque, Aligned and MisIndexed. World labels are system-specific; c and d reuse the same crystallization campaigns. Each column contains fifteen pairs of independent sessions. Scales differ, and larger batch budgets also provide larger operational and computational allowances. Complete budget readouts, including electrochemical optimization and fines coverage, are in Supplementary Fig. S3.

Independent sessions can use the two research envelopes in different ways. A selected crystallization pair illustrates early feasibility followed by refinement in the smaller session, and a late transition to feasibility in the larger session; its operations and retests are shown in Fig. 2. The complete six-outcome budget comparison is retained in Supplementary Fig. S3.

The positive electrochemical and partitioning results show that the agent can turn a larger research envelope into better forecasts. The crystallization results show that this benefit does not extend uniformly to every response or to delivery. These comparisons neither isolate the value of twelve additional observations nor rule out missing discriminating evidence. Alongside resources, supplied scientific information can also change the path from experimental choice to prediction.

Prior information helps within some regimes but fails across others

Prior information has demonstrable benefits. In the reaction parameter study, aligned information lowers discovery macro MAE from 0.10186 to 0.05701 and optimization error from 0.13893 to 0.08384, beating Opaque in four of five worlds under each commission (Supplementary Fig. S2a,b). Across other blocks, however, there is no common ordering of the three arms. Close aggregate means in the equilibrium entity and structural studies are not evidence of equivalence.

The equilibrium parameter study gives a sharper pattern. Overall macro MAE is 0.01214 for Opaque, 0.04314 for Aligned and 0.03928 for MisIndexed. Opaque outperforms both dossier conditions in all five worlds. A post hoc grouping by public recipe concentration locates the reversal (Table 1). On the nine questions other than the three most dilute, Aligned improves MAE from 0.01008 to 0.00465. On the three most dilute, its MAE instead rises from 0.01833 to 0.15862. Both directions occur in every world. The remaining nine questions include boundary conditions and should not all be called interpolation.

Aligned intervals also fail selectively: coverage is 91.3% on the other nine questions but 2.2% on the dilute three, versus 90.7% and 93.3% for Opaque. The dilute recipes have nominal concentrations near 10^{-4} to 10^{-5} M; all fifteen source campaigns remain above 0.018 M. Their measured dissociation occupies a low-response plateau, while the dilute tests require leaving it.

Table 1. Equilibrium prediction performance across test regimes.

Information arm	Other nine: MAE	Dilute three: MAE	Other nine: coverage	Dilute three: coverage
Opaque	0.01008	0.01833	90.7%	93.3%
Aligned	0.00465	0.15862	91.3%	2.2%
MisIndexed	0.00439	0.14397	89.6%	13.8%

Values average five worlds per arm and three response targets: normalized pH, dissociation fraction and precipitation signal. Coverage concerns the original 80% prediction intervals, with 675 reference-observation judgments per arm for the other nine queries and 225 for the dilute three; these are not independent world replicates. The grouping is exploratory. The single-query dissociation comparison in Fig. 5 has a different scope.

The retained trajectories connect these outcomes to the original research process. Across all 180 source assays, dissociation ranges from 5.5% to 9.2%. At the most dilute test recipe, reference means instead range from 44.8% to 71.6%; Opaque predictions follow this increase, whereas Aligned remains near the observed plateau (Fig. 5). In one world, Aligned explores a ninety-fold concentration range and Opaque a fourfold range, yet neither directly samples the dilute regime. Broad input coverage therefore need not provide evidence distinguishing behaviour across response regimes.

We examined the public prediction explanations from all fifteen original campaigns. Four Opaque accounts explicitly use piecewise weak-acid extrapolation and the fifth permits a qualitative departure from the plateau. All five Aligned accounts primarily extend a plateau or weak local trend. MisIndexed includes a counterexample: one campaign mixes weak-acid and plateau explanations and predicts appreciable dissociation at the most dilute condition. These retrospective descriptions connect numerical performance to the agent’s expressed prediction rules, without establishing that supplied information caused a particular internal reasoning mechanism.

The chronology matters. In one Opaque campaign, K1 already estimates an effective acid constant and an active-pool cap, and Q carries that hypothesis into low-concentration predictions. In another, K1 limits its account to the measured plateau; Q first makes the unsaturated weak-acid extrapolation explicit. Its aligned counterpart likewise acknowledges in K1 that the plateau need not extend to zero loading, but predicts a dilute dissociation fraction of 0.0804 [0.053, 0.108], against 0.629 [0.45, 0.80] for Opaque and a reference mean of 0.65463. Predictive success belongs to the complete research-and-assessment sequence; it cannot all be credited to knowledge already sealed in K1.

The equilibrium contrast concerns the entire autonomous process: information can influence both the evidence acquired and the way it is used. The observed pattern does not identify prior anchoring as its cause. It instead raises a complementary question: when a relationship does remain stable under the withheld interventions, do the agent’s predictions preserve it?

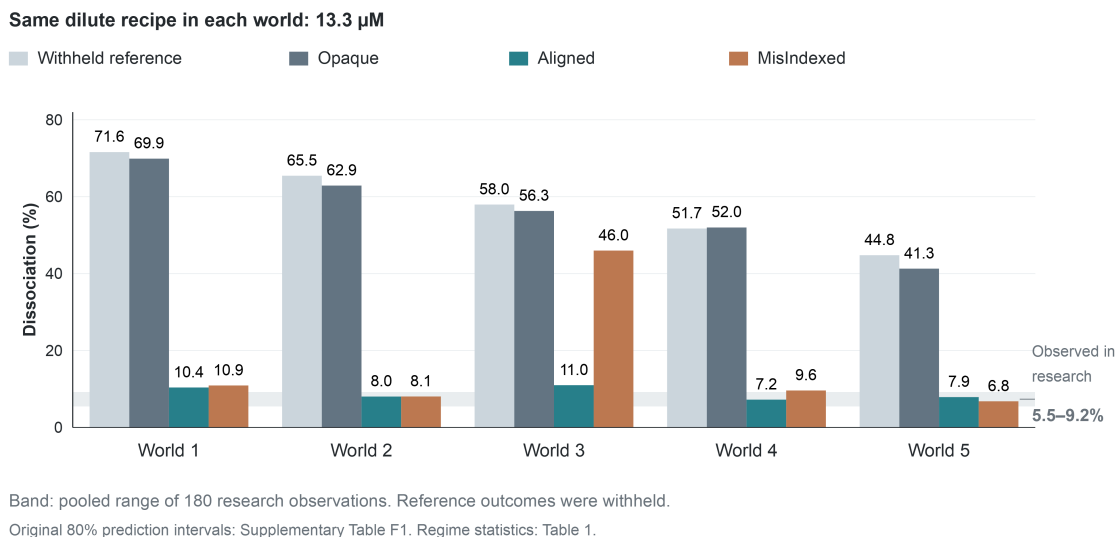


Figure 5: Original predictions and withheld outcomes at the most dilute equilibrium test. Bars show the original five-observation reference mean and each agent’s point prediction for the same recipe within each of five worlds, at a nominal concentration of 13.3 micromolar. The pale horizontal band is the pooled minimum-to-maximum dissociation range of all 180 source assays from fifteen campaigns; it is not an uncertainty interval or a world-specific range. All reference outcomes were withheld from the agents. The MisIndexed World 3 counterexample is retained. Complete original 80% prediction intervals and campaign-specific source ranges are reported in Supplementary Table F1. This single-response, single-query comparison is distinct from the three-response regime aggregates in Table 1.

Generalization can fail by both preserving and revising the wrong relationships

Crystallization provides the complementary problem: do forecasts preserve an observed response when it remains stable under new conditions? Source and withheld purity average 98.53% and 98.51%, respectively, whereas agent predictions average 93.54% at 12 batches and 94.89% at 24 (Fig. 6a). Of 360 query forecasts, 335 are too low. The mean within-world withheld-purity span is only 2.442 percentage points. This is stability within the tested conditions, not a claim that purity is invariant under arbitrary interventions.

For each campaign, we construct a baseline that predicts the mean of its own public final assays for every withheld recipe. It acquires no additional experiments and uses no withheld outcomes. Agent recovery forecasts outperform this baseline in 26 of 30 campaigns and fines forecasts in 21, whereas size forecasts win in four and purity forecasts in none (Fig. 6b). Mean recovery MAE is 9.18 percentage points for the agent versus 17.14 for the baseline; purity MAE is 4.35 versus 0.48. These response-specific results argue against interpreting the purity error as a general inability to use data. The narrow withheld-purity range explains the baseline’s strength without establishing mechanism discovery. Full error magnitudes and a nearest-neighbour reference using public recipe features are reported in Supplementary Table E1.

The same information condition can help one response and hurt another (Fig. 6c). At 12 batches, both Aligned and MisIndexed improve recovery MAE while worsening purity MAE relative to Opaque in every crystallization world, including all four both-conforming pairs for each comparison. At 24 batches, the joint direction occurs in only two and three of five worlds, respectively. The contrast is therefore neither a universal penalty for supplied information nor a budget-invariant law.

Reported uncertainty does not consistently accommodate the purity errors (Fig. 6d). At 12 batches, nominal 80% interval coverage is 98.3% for Opaque, 60.0% for Aligned and 23.3% for MisIndexed; at 24 batches it is 95.0%, 73.3% and 55.0%. Aligned and MisIndexed intervals are wider on average than Opaque intervals at both budgets, so their poorer coverage cannot be described simply as narrower reported uncertainty. Conversely, Opaque’s high coverage alone does not establish calibrated or sharp intervals (Supplementary Table E2).

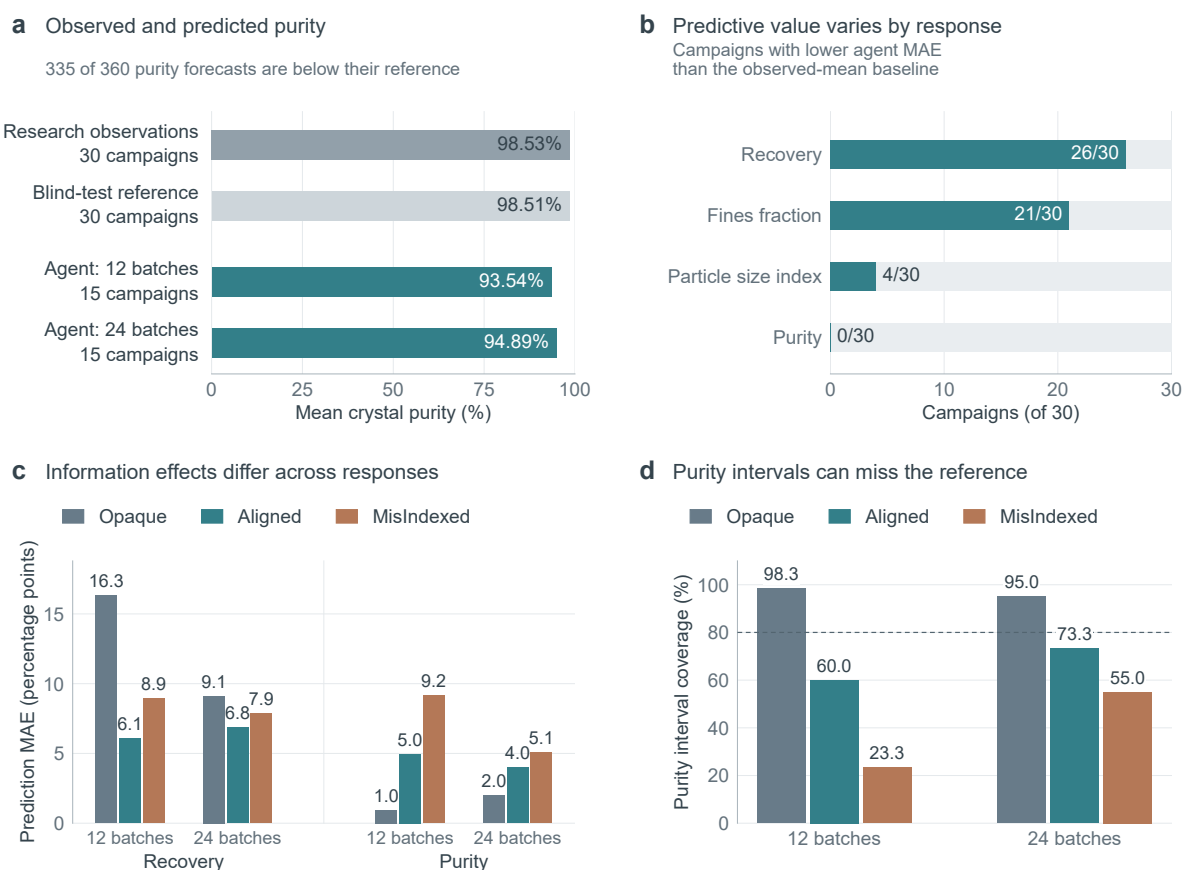


Figure 6: Crystallization forecasts show response-specific departures from experimental evidence. **a**, Mean purity in source observations and withheld reference outcomes (thirty campaigns each), compared with forecasts at each budget (fifteen campaigns each); bars equally weight campaign means. **b**, Campaigns with lower agent MAE than a predictor repeating the same campaign’s public final-assay mean for every withheld recipe. Grey remainders favour the baseline; there are no ties. Comparisons are within each response, not between response scales. **c**, Recovery and purity MAE by information arm and budget, with five worlds per bar. **d**, Coverage of original nominal 80% purity intervals, averaged over five campaigns of twelve queries per arm and budget. The dashed line marks the nominal level. Five worlds recur across arms and independent budget sessions; queries are not independent world replicates. All thirty campaigns, including the source-assay shortfall, remain included. Bars are descriptive summaries, not confidence intervals. The baseline uses no additional experiments or withheld outcomes.

The equilibrium and crystallization findings expose opposite errors in using an observed relationship (Figs. 5 and 6). Aligned equilibrium forecasts extend a plateau beyond its applicable regime. Crystallization forecasts depart from a response that remains approximately stable over the tested conditions. In both cases the world’s process laws remain fixed; the intervention changes the conditions under which those laws generate a response. Preservation and revision here describe forecasts relative to the source observations, not directly measured changes in internal belief.

The dilute-equilibrium coverage failure and the response-specific crystallization intervals make uncertainty part of this applicability problem. A selected crystallization account further illustrates that acknowledging limited evidence need not yield calibrated predictions (Supplementary Information, section E.2).

The selected crystallization pair also distinguishes sealed prediction from retrospective reflection (Supplementary Fig. S4). Both sessions forecast the wrong direction for a thermal intervention, then acknowledge limitations in K2 without receiving reference feedback. Their proposed follow-up remains unexecuted. A useful retrospective explanation therefore remains distinct from a correct sealed prediction.

Together, these patterns motivate checking not only which relationships a researcher expresses, but where its predictions treat those relationships as applicable. Successful Opaque equilibrium extrapolations, the mixed MisIndexed account and the recovery advantage over the empirical mean remain counterexamples to uniform failure. These observations do not establish a common internal cause.

Discussion

ChemWorld connects controlled process models to the open sequence of decisions made during autonomous research. Its empirical contribution is that operational achievement, predictive generalization and warranted uncertainty can diverge, with benefits from resources and prior information depending on the response and intervention regime. Positive partitioning and reaction results are essential: the study includes successful uses of experimental evidence as well as errors. The equilibrium and crystallization findings make the applicability of acquired relationships a concrete evaluation problem.

Considering the two patterns together rules out a simple methodological prescription from either example alone. Copying the observed mean works unusually well for crystallization purity but fails badly on dilute equilibrium tests. Mechanistic extrapolation supports successful Opaque predictions in equilibrium, yet a plausible explanation is not itself a guarantee of predictive accuracy. What matters is how experimental evidence constrains the conditions over which a proposed relationship is used, including uncertainty about alternatives compatible with the observed data.

The comparisons assess complete research processes and do not isolate the causal contributions of evidence acquisition and interpretation. Larger resource envelopes and correct partial information do not establish that discriminating evidence was sufficient. Full-context prediction does not test report-compression loss, and retrospective public explanations do not reveal internal cognitive causes. The evidence is limited to one agent configuration, five reused world instances per study, correlated readouts and some post hoc analyses. The finite model domain, absence of wet-laboratory validation and protocol differences across blocks constrain broader claims. These limits apply equally to the observed successes and failures.

Evaluation of autonomous science should therefore connect what a researcher accomplishes to what its experiments support: the relationships it forms, the conditions under which its forecasts preserve or change them, and the uncertainty it assigns to those forecasts. Retaining the research history and the timing of reports and predictions allows those connections to be checked without replacing the original researcher with a separate record-reading task. Scientific generalization requires more than a useful recipe or a plausible account; it requires predictions whose claimed scope is tested by new interventions.

Methods

World construction and execution

A world is specified by public experimental contracts and private process parameters. Process components declare state, domains, dependencies and compatible interfaces. Operation and observation models compile these declarations into a laboratory interface. The participant can act through that interface but cannot inspect the simulator, private parameters, evaluator records or repository. Instruments may consume samples and operations consume declared material, time and action allowances. Candidate transitions are validated before commitment; rejection preserves committed physical state while retaining the declared attempt consequences. Replay concerns the environment and recorded operations, not regeneration of language-model responses.

Platform qualification

The frozen platform qualification passed 64 task-world units, 1,786 boundary and categorical recipes, 192 invalid-action probes and 52 generated compositions. The generated cases comprise 18 previously absent topologies, eight new task-world identities using a registered topology and 26 additional coverage cases. Component and interface checks passed 32 module probes, seven interface paths and seven invalid compositions. These finite-domain checks establish internal model and execution consistency; they do not establish agreement with arbitrary wet-laboratory chemistry. Exact environment replay reconstructs executed actions and resource consequences, including rejection and rollback.

Qualification distinguishes internal physical and accounting consistency from successful research. It checks executable domains, resources, information separation, valid query references and exact replay. Invalid actions and incompatible compositions have explicit expected rejection or rollback behaviour. Subsequent study adapters have their own checks; the frozen platform qualification is not an automatic certificate for later execution code. The reported autonomous campaigns retain their development-evidence status and are not promoted to formal validation by this analysis.

Study units and objectives

The eight blocks comprise electrochemistry with entity priors (60 campaigns), partitioning with entity priors (30), reaction processing with parameter and structural priors (60), equilibrium with entity, parameter and structural priors (15 each), crystallization with entity priors (30) and purification with entity priors (15). Every block uses five worlds and three information arms. Electrochemistry and reaction processing have discovery and optimization commissions. Electrochemistry, partitioning and crystallization have 12- and 24-batch conditions; other blocks use 12. Parameter-prior P and purification P denote different concepts.

Electrochemistry varies materials, potential, current limits and duration; its optimization objective is a public balanced-efficiency score. Reaction processing varies materials, catalyst loading and thermal histories. Partitioning studies allocation under material, volume and process changes. Equilibrium studies amount, volume, concentration, staged additions, dissociation and precipitation. Crystallization seeks recovery net of seed mass subject to purity at least 0.80, fines fraction at most 0.50 and particles present. Purification seeks original-charge recovery subject to purity at least 0.80. Recovery denominators differ and are not combined.

Agent and information conditions

All source campaigns use GPT-5.6 Sol with medium reasoning effort in independent sessions. Within a block, arms share the physical world, operations, resource contract and withheld questions. The treatment modifies the public dossier according to the block's fixed entity, parameter or structural design. Opaque

still exposes the operational interface and general task background. Aligned is a partial nominal description, such as a true effective-pKa interval in the equilibrium parameter study. MisIndexed applies the specified mismatch rather than supplying arbitrary misinformation.

Agents may choose conditions, repetitions, measurement allocations and experiment grouping, and may use the bounded public numerical tool. They are not required to choose from mechanism templates, emit a fixed equation or reflect after each batch. Batches, operations, provider responses and decisions made after feedback are different counts. Each budget permits a matching number of additional measurements and a final assay per completed batch; other resources remain separately bounded. The larger budget includes a larger execution and computation envelope. Retained early electrochemistry and partitioning assessments have an eight-call calculator allowance; later blocks disclose up to 128 attempts. Outputs are requested in English, but retained reaction and equilibrium assessment prompts include Chinese instructions requesting English responses. These differences remain part of the historical protocol.

Sealed assessments and retests

Where required, a completed-batch recommendation is sealed before assessment and its procedure retested independently with a different observation seed in the same world. There are 165 such retests. K1 requests an unrestricted scientific account including relationships, evidence, applicability and unresolved alternatives. Q then requests point predictions and intervals for twelve fixed new-batch recipes without laboratory access or target feedback. K2 follows the sealed predictions with retrospective questions about support, limitations and discriminating experiments. The short version has three questions; reaction and equilibrium use seven themes. The bounded equilibrium parameter block adds fifteen structured supplements.

The original research context is available during every stage. Q can use the source history in addition to K1, and K2 cannot revise Q. Consequently, neither report-only knowledge transfer nor a uniform mechanism-correctness score is an endpoint. The bounded equilibrium review described below covers all fifteen campaigns in that block; complete mechanism annotation across the 240-campaign programme has not been performed.

Outcome definitions and references

For each response, campaign MAE averages absolute point error across the twelve questions. Interval coverage measures the fraction of specified reference targets contained in the reported intervals; width is their average span. Partitioning uses nominal 90% intervals and the other blocks use 80%. Partitioning and crystallization point references use underlying pre-final-assay response values. Electrochemistry and purification use fixed seeded observations. Reaction and equilibrium point references average five observations; their coverage is evaluated against the individual observations. Macro MAE averages response-specific MAEs equally within the designated block. These targets and scales are not pooled across systems.

The crystallization mean baseline uses only the source campaign's public final assays. Nearest-neighbour prediction additionally uses public recipe features. A parser correction excludes resource-denied attempts from physical recipe features while preserving those attempts in the record. It restores two baseline evaluations; the other twenty-eight remain exactly unchanged. No agent or simulator is rerun for the correction. The equilibrium source-mean diagnostic is a separate post hoc reference. Its failure on dilute queries confirms that copying the plateau is insufficient for those questions; it is not a strong system-identification comparator.

Descriptive comparisons and retained failures

Goal comparisons match world, information arm, budget and prior locus. Budget comparisons match world, arm and goal, comparing independent sessions. Arm comparisons match the other factors within each study. Means weight campaigns equally. Five world clusters recur across arms, goals and budgets; questions,

response metrics and batches are correlated observations. We report magnitudes, paired directions, all-arm tables and selected sensitivity analyses rather than inferential tests treating queries as independent trials.

The equilibrium dilution grouping was selected post hoc using the public concentrations of the three most dilute recipes, approximately 1.33×10^{-4} , 1.33×10^{-5} and 1.67×10^{-4} M. Grouped scores reproduce each original twelve-question macro score when weighted 3:9. The joint recovery–purity crystallization comparison is also exploratory. Neither changes the original query bank or replaces an unfavourable outcome. Retrospectively selected trace examples are distinguished from cohort-level findings.

All 240 campaigns remain included. One crystallization source has eleven rather than twelve final assays after solvent exhaustion. One purification source consumes twelve starts but has ten final assays and two discarded vessels. Both are marked nonconforming; neither is replaced. Infrastructure retries retain previous attempts and resource accounting. The supplementary tables report the complete matrix, targets, qualification domains and every response-specific arm mean. Original protocol variation and excluded historical pilots are documented separately from this pool.

Retrospective analysis of the autonomous equilibrium process

We include all fifteen effective parameter-prior equilibrium sources from their retained source and trajectory indexes: five worlds, three arms, 180 batches, 945 operations and 45 canonical assessment stages. Nominal batch concentration is the sum of reagent additions divided by the sum of solvent additions; it is not inferred dissolved concentration. Original question scores and reference observations remain unchanged. No additional model, simulator or judge calls are used.

A single author review assigns descriptive categories to the public Q explanation: piecewise weak-acid extrapolation, qualitative departure from the plateau, plateau or weak-trend extension, and mixed weak-acid/plateau explanation. Exact excerpts and prediction values accompany the classifications; they are not an independent mechanism assessment. K1 evidence is distinguished from hypotheses first explicit in Q. K2 future-experiment choices are classified separately: eleven of fifteen propose trace-load experiments after seeing Q conditions, without receiving reference feedback. These proposals were not executed and do not establish spontaneous discovery or repair during the source campaign. The exported decision-reason fields are unavailable for all 945 operations, so no contemporaneous reasoning sequence is reconstructed from them.

Data and code availability

The source observations, sealed outputs, evaluator-derived summaries and analysis code underlying this draft are retained in the project evidence records. The supplementary material reports aggregate outcomes and protocol details. A public archival package for this integrated study, including the applicable execution version and privacy-reviewed records, has not yet been finalized; the earlier platform release should not be assumed to contain these later studies.

References

- [1] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624:570–578, 2023. doi: 10.1038/s41586-023-06792-0.
- [2] Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6:525–535, 2024. doi: 10.1038/s42256-024-00832-8.
- [3] Nathan J. Szymanski et al. An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature*, 624:86–91, 2023. doi: 10.1038/s41586-023-06734-w.
- [4] Kobi C. Felton, Jan G. Rittig, and Alexei A. Lapkin. Summit: Benchmarking machine learning methods for reaction optimisation. *Chemistry–Methods*, 1:116–122, 2021. doi: 10.1002/cmtd.202000051.
- [5] Florian Häse, Matteo Aldeghi, Riley J. Hickman, Loïc M. Roch, Melodie Christensen, Elena Liles, Jason E. Hein, and Alán Aspuru-Guzik. Olympus: a benchmarking framework for noisy optimization and experiment planning. *Machine Learning: Science and Technology*, 2:035021, 2021. doi: 10.1088/2632-2153/abedc8.
- [6] Chris Beeler, Sriram Ganapathi Subramanian, Kyle Sprague, et al. ChemGymRL: A customizable interactive framework for reinforcement learning for digital chemistry. *Digital Discovery*, 3:742–758, 2024. doi: 10.1039/D3DD00183K.
- [7] Peter Jansen, Marc-Alexandre Côté, Tushar Khot, Erin Bransom, Bhavana Dalvi Mishra, Bodhisattwa Prasad Majumder, Oyvind Tafjord, and Peter Clark. DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0324.
- [8] Kanishk Gandhi, Michael Y. Li, Lyle Goodyear, Agam Bhatia, Louise Li, Aditi Bhaskar, Mohammed Zaman, and Noah D. Goodman. BoxingGym: Benchmarking progress in automated experimental design and model discovery, 2025. Preprint, arXiv:2501.01540.
- [9] Adam N. Elmachet and Paul Grigas. Smart Predict, then Optimize. *Management Science*, 68(1): 9–26, 2022. doi: 10.1287/mnsc.2020.3922.

Supplementary information

A Protocol detail and evidence handling

A.1 Complete study matrix

System / prior locus	Scientific commission	Budgets	Campaigns	Final-assayed batches
EC / E: electrochemistry	Discovery; score optimization	12, 24	60	1,080 / 1,080
PA / E: phase partitioning	Explain phase allocation	12, 24	30	540 / 540
RX / P and S: reaction and thermal processing	Discovery; safety-constrained optimization	12	60	720 / 720
EQ / E: medium identity	Characterize entity-conditioned responses	12	15	180 / 180
EQ / P: bounded equilibrium	Characterize effective responses	12	15	180 / 180
EQ / S: structural equilibrium	Explain response structure	12	15	180 / 180
C / E: crystallization	Deliver crystals under quality constraints	12, 24	30	539 / 540
P / E: purification	Recover product subject to purity	12	15	178 / 180
Total	Six system families		240	3,597 / 3,600

Table A1. Complete selected study scope. Each campaign has one agent realization. One C campaign exhausts its solvent stock and discards its twelfth vessel, completing eleven final assays. One P campaign uses all twelve vessel starts but discards two vessels, leaving ten final assays. All campaigns have sealed mechanism, prediction, and reflection stages; 238 meet the full source-and-assessment specification.

A.2 Prediction targets and system-specific questions

System	Prediction responses	Reference target	Interval level
EC	Six conversion/efficiency responses, including score	Seeded final observation	80%
PA	Organic and aqueous product fractions	Noiseless pre-sampling fractions	90%
RX	Yield, conversion, selectivity, byproduct signal, risk, score	Five-observation mean; coverage over observations	80%
EQ / E, P and S	Normalized pH, dissociation, precipitation signal	Five-observation mean; coverage over observations	80%
C	Net recovery, purity, size index, fines fraction	Noiseless pre-assay responses	80%
P	Purity, original-charge recovery	Seeded final observation	80%

Table A2. Every campaign predicts twelve conditions. PA fractions are complementary, and its two decision questions reuse the conditions. Reference repeats reduce observation noise but do not add agent sessions.

EC’s twelve conditions balance positive and negative potential while varying duration, materials, and current cap at a stated loading. Its six responses are selective product yield, electrochemical selectivity, Faradaic efficiency, transport efficiency, energy efficiency, and public score. Current is a magnitude cap, not guaranteed delivered current. PA predicts phase fractions before analytical sampling; sampling losses and complementary fractions must not be counted as separate discoveries.

RX’s twelve paths examine time, temperature, catalyst, heating sequence, and quenching. Actual thermal telemetry matters: requested boundary temperature need not be realized reaction temperature. Five reference observations per condition are withheld from agents, with all source assessments sealed before reference-truth release. RX’s 600 reference executions are separate from 720 source batches and 60 retests.

Each EQ block uses twelve questions and five observations per question, giving 300 reference executions. EQ/E crosses three medium identities with three concentrations and a matched-concentration scale control, assessing entity contrast and concentration and scale transfer under a fixed common topology. Bounded EQ/P examines effective aqueous relationships; canonical EQ/S supports response-shape analysis without requiring a named hidden family. An instrument’s equilibrium diagnostic is not the agent’s subjective confidence. The bounded study’s effective-parameter supplement is additional to K1/Q/K2 and does not confer discovery credit for a parameter already supplied as a treatment.

C’s twelve conditions form six pairs: material, seed dose, cooling history, prior thermal history, continued growth, and upstream loading. Its size response is $\min(d_{50}/250 \mu\text{m}, 1)$, a bounded number-weighted index rather than a diameter in micrometres. Fines are particles below 20 micrometres. Quality requires purity and fines criteria jointly, with particles present; recovery of 0.10 is an initial feasibility target, not an optimization ceiling. P’s six paired interventions concern multistage recovery, including wash staging. Its original-charge denominator differs from C’s crystallization-stage recovery.

RX and EQ coverage uses individual reference observations while point targets use means. EC and P use single seeded observations. PA and C prediction coverage concerns noiseless pre-sampling responses; C recommendation retests use observed assays. Apparent retest threshold crossings can therefore reflect noise. Macro errors weight a system’s responses equally; normalized scales do not make responses physically or scientifically interchangeable.

A.3 Language, numerical tools, and stage budgets

Source briefs and requested reports are English. EC/PA and C/P assessment instructions are English; RX and EQ K1/Q/K2 instructions include Chinese text requesting English reports. Within-block arms have the same language contract. Seven-theme K2 reviews cover treatment claims, influential experiments, competing explanations, one proposed discriminating experiment, goal-related tradeoffs, unused evidence and uncertainty, and limitations of the recommendation or scientific account. Three-question versions combine these themes. The reviews occur after source experiments and sealed predictions; their length does not provide additional source experiments.

The earlier EC/PA block retains its saved eight-call diagnostic-calculator hard stop; later blocks disclose 128-attempt assessment allowances. An old EC parameter/structural supplement encountered the smaller limit and remains an execution failure outside the primary E cohort. The repair is not retroactively attributed to older sessions. Public calculation uses agent-accessible information, not hidden targets. Differences in numerical budgets and deadlines are execution contracts, not scientific findings.

A.4 Completion, recovery, and exclusions

The primary denominator is 240 scheduled campaigns, not provider attempts. All 720 K1/Q/K2 stages and 165 planned recommendation retests in EC, RX, C, and P are complete, plus fifteen EQ supplements. Final-assayed source batches total 3,597 of 3,600 planned. C's second-world 12-batch Opaque source has eleven final assays. P's fifth-world Opaque source starts twelve vessels, discards two, and completes ten final assays. Both remain in outcome tables with conformance indicators.

EC/PA has 68 first-attempt and 22 infrastructure-recovered campaigns. Recovery after transport or host failures retains earlier attempts and consumed resources. Missing assessment stages resume from saved context when the source is intact; fresh source replacements are recorded as such. Repeated question delivery and failed-attempt computation are not silently counted as an original clean attempt. EC goal sensitivity is reported in the main text. First-attempt-only EC-discovery, EC-optimization, and PA budget comparisons improve prediction in four of five, eight of eleven, and eight of ten pairs, respectively; these reduced subsets have uneven coverage.

The current P matrix follows a correction of prepartition inventory handling. Affected qualification and the entire agent block were restarted under the corrected contract; earlier apparently successful cells were not selected into the new matrix. All current source and recommendation trajectories pass inventory, resource, and replay checks. Its two lawful discards are a source-assay shortfall, not an inventory failure. The fixed purity-direction rule treats changes of 0.02 or less as small; a numerical tolerance of 10^{-12} corrects floating-point equality handling. Six direction classifications changed after this correction, with originals preserved. Predictions, point errors, intervals, retests, and denominators were unchanged.

Historical single-world EC parameter/structural supplements, the closed-set EQ/S pilot, early C/P development runs, qualification recipes, references, replays, and interrupted attempts do not enlarge the denominator. Their records remain separate. Frozen platform qualification is also a separate evidence programme. These counts describe execution and analysis scope; they are not 240 independent chemistry replications.

A.5 Baseline correction and trace interpretation

The C empirical references use the same source final observations for all three arms. A recipe parser originally rejected resource-denial transaction records in two sources. The correction preserves those attempts as provenance and excludes them from committed physical recipe features. Recomputing all thirty baselines restores the two missing evaluations and exactly reproduces the twenty-eight previously available results. Original trajectories, truth, model answers, and error records are retained; the correction uses no new simulator or provider calls. Unknown transaction statuses remain errors. This is an evaluator repair, not a change to a completed experiment.

The EC illustrative trace is the second-world, 12-batch Opaque EC discovery campaign; its sparse-positive-coverage counterexamples are first-world, 24-batch Aligned and MisIndexed discovery campaigns. They are retrospectively selected examples, not a coded prevalence estimate. The additional C case is the fifth-world, 12-batch MisIndexed campaign, chosen as the largest purity MAE in the 12-batch cohort. Its extreme error illustrates a failure; it is not a representative sampling rule.

A systematic analysis should attach each mechanistic claim to its experimental support, contradictory observations, stated domain, and corresponding predictions. It should distinguish an unperformed discriminating experiment, acquired evidence omitted from an account, justified unresolved alternatives, and report-prediction inconsistency. Evaluator knowledge helps identify discriminating interventions; a claim is not wrong merely for using an equivalent representation. Independent review can assist this analysis without replacing measured predictions and retests. No completed cohort-wide mechanism score or causal compression experiment is implied.

A.6 Full prior overview

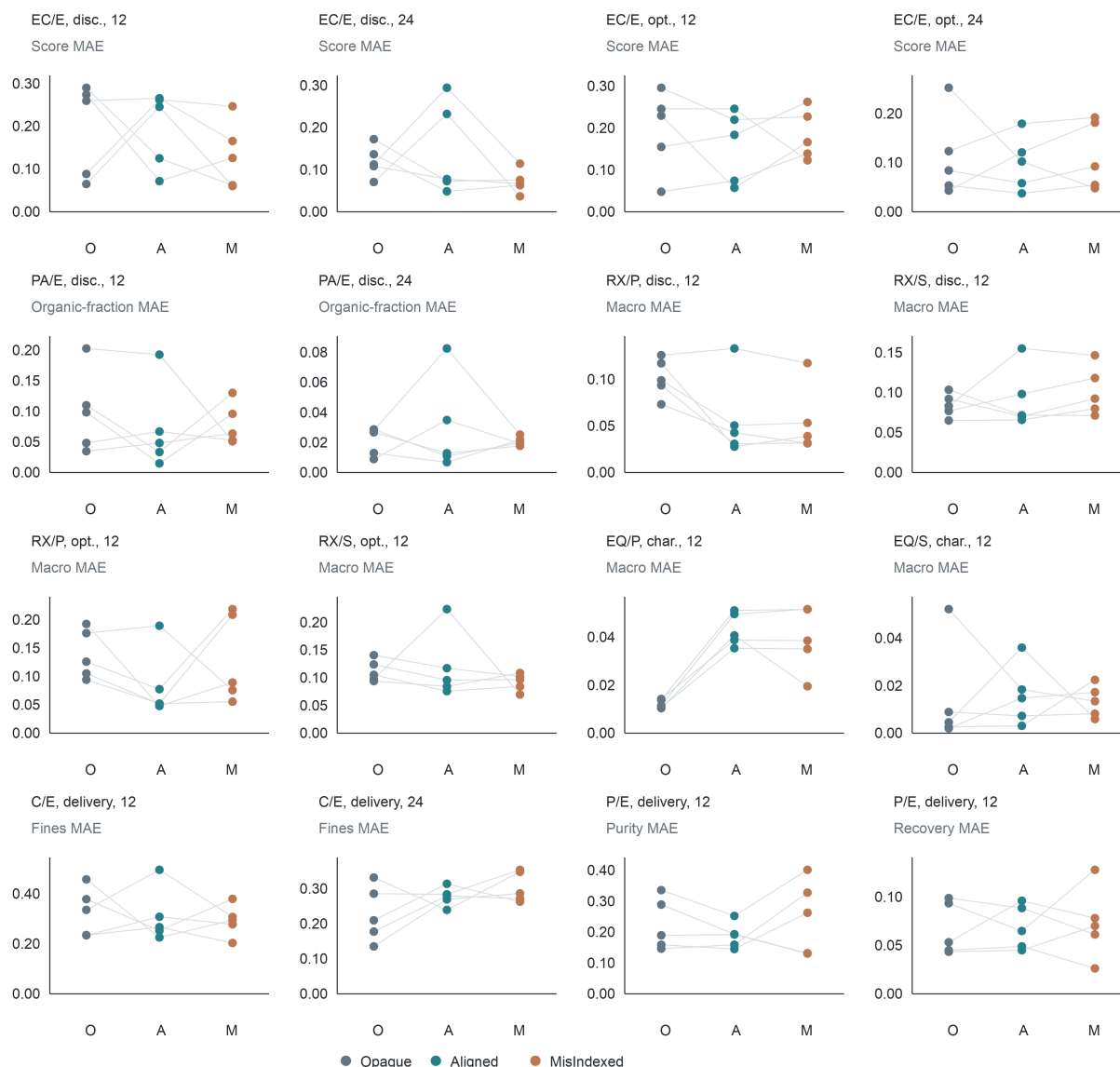


Figure S1: Prior information has no common ordering. Each panel retains five worlds and three information conditions: Opaque (O), Aligned (A) and MisIndexed (M). Grey lines connect the same world across arms; colored points show individual campaigns, including source-assay shortfalls. RX and EQ use within-system macro errors; other panels name the response. Scales differ. The complete response tables include the equilibrium entity-prior study and all 240 campaigns.

A.7 Prior effects by system and regime

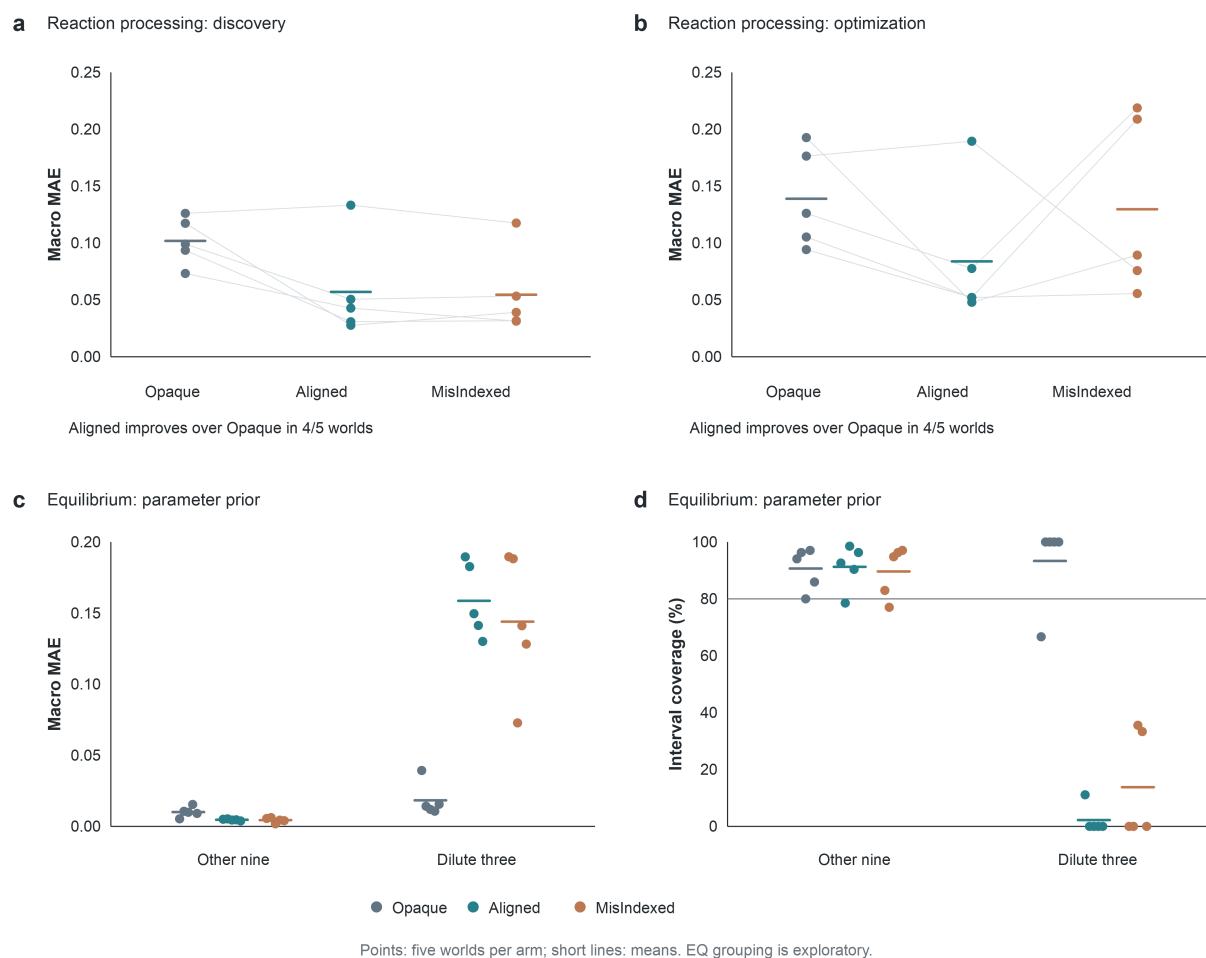


Figure S2: Prior information depends on the system and intervention regime. **a,b**, Reaction parameter-prior macro MAE under discovery and optimization; lines connect the same world across arms. **c,d**, Equilibrium macro MAE and interval coverage on the other nine and three most dilute queries. Points retain five worlds per arm; short lines show means. The horizontal reference marks nominal 80% coverage. The exploratory grouping does not classify all other nine queries as interpolation.

A.8 Complete budget comparison

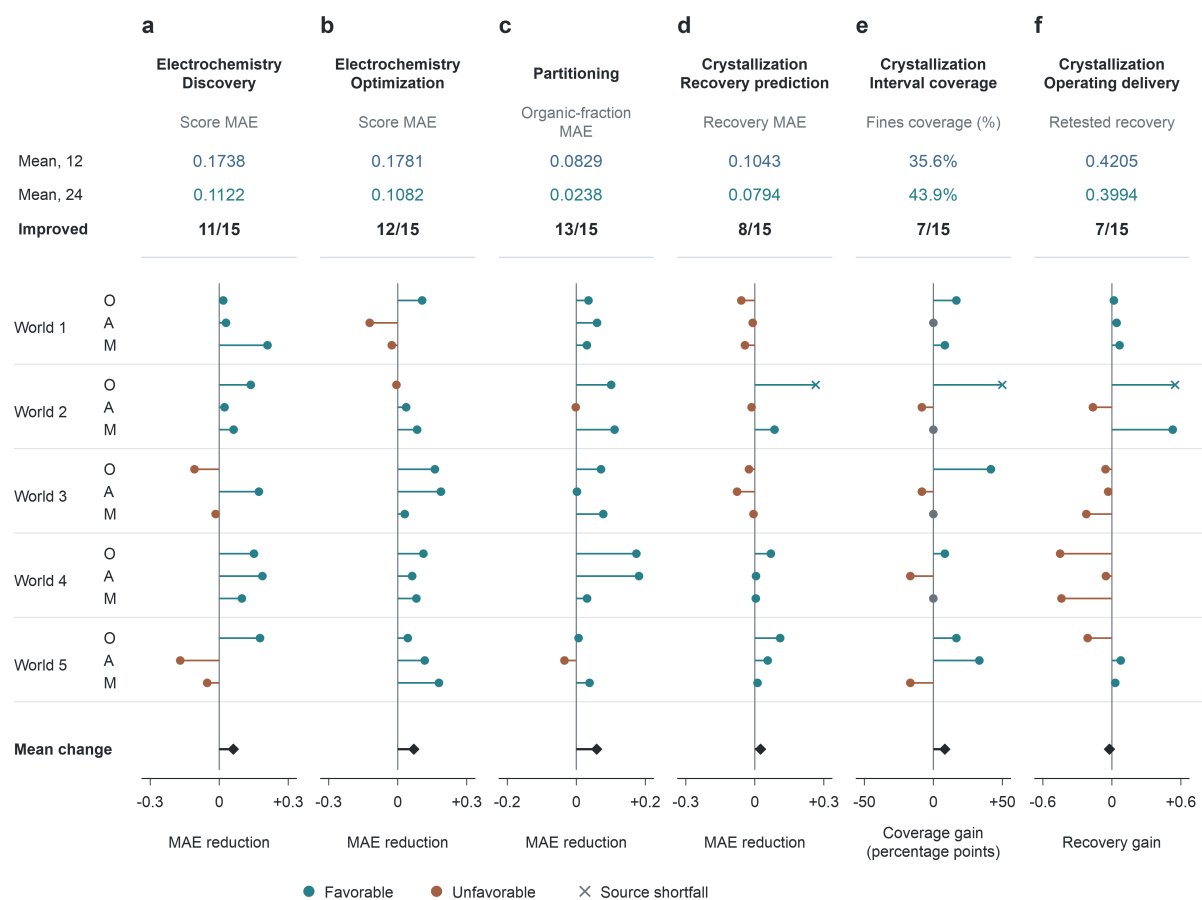


Figure S3: Complete budget effects across six readouts. Each column retains fifteen pairs of independent sessions. Headers show campaign means and improved counts; rows show paired changes. Positive changes favour the larger research envelope. Panels a-d show MAE at 12 minus MAE at 24; e shows fines coverage at 24 minus coverage at 12 in percentage points; f shows the corresponding gain in retested recovery. All fines coverage values remain below nominal 80%. Diamonds show mean changes; crosses retain source-assay shortfalls. O, A and M denote Opaque, Aligned and MisIndexed. World labels are system-specific. Scales differ except in a-b. The crystallization panels reuse the same campaigns; larger budgets also increase computational allowances.

B Complete metric-specific arm means

Each entry is mean absolute error / empirical interval coverage (%), averaged over five world sessions. All scheduled sessions are retained, including the two source-budget shortfalls. PA uses nominal 90% intervals; all other studies use 80%. Metric scales and reference targets differ across systems. The machine-readable table includes each campaign’s interval width and protocol status.

B.1 EC, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2983 / 55.0	0.2740 / 65.0	0.1272 / 88.3
Energy efficiency	0.3098 / 33.3	0.2595 / 51.7	0.1582 / 63.3
Faradaic efficiency	0.2580 / 46.7	0.2259 / 60.0	0.1356 / 73.3
Public score	0.1953 / 51.7	0.1939 / 60.0	0.1321 / 71.7
Selective product yield	0.0731 / 60.0	0.0817 / 38.3	0.0598 / 56.7
Transport efficiency	0.2602 / 45.0	0.2184 / 66.7	0.1328 / 76.7

B.2 EC, E prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2908 / 63.3	0.2378 / 66.7	0.2661 / 73.3
Energy efficiency	0.2794 / 43.3	0.2541 / 31.7	0.2440 / 55.0
Faradaic efficiency	0.2506 / 53.3	0.2532 / 31.7	0.2315 / 56.7
Public score	0.1947 / 55.0	0.1561 / 58.3	0.1835 / 53.3
Selective product yield	0.0758 / 43.3	0.0623 / 53.3	0.0762 / 50.0
Transport efficiency	0.2478 / 58.3	0.2053 / 51.7	0.2217 / 60.0

B.3 PA, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3
Organic fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3

B.4 EC, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1902 / 68.3	0.1268 / 80.0	0.0428 / 93.3
Energy efficiency	0.1581 / 61.7	0.1651 / 68.3	0.0945 / 80.0
Faradaic efficiency	0.1458 / 65.0	0.1505 / 66.7	0.0877 / 78.3
Public score	0.1201 / 63.3	0.1452 / 73.3	0.0713 / 78.3
Selective product yield	0.0726 / 53.3	0.0573 / 70.0	0.0374 / 75.0
Transport efficiency	0.1495 / 65.0	0.1501 / 66.7	0.0846 / 78.3

B.5 EC, E prior: optimization, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1403 / 85.0	0.1252 / 85.0	0.0864 / 90.0
Energy efficiency	0.1977 / 48.3	0.1307 / 60.0	0.1779 / 65.0
Faradaic efficiency	0.1502 / 76.7	0.1098 / 78.3	0.1515 / 71.7
Public score	0.1113 / 75.0	0.0996 / 71.7	0.1136 / 75.0
Selective product yield	0.0515 / 75.0	0.0498 / 56.7	0.0587 / 58.3
Transport efficiency	0.1420 / 80.0	0.1257 / 73.3	0.1570 / 68.3

B.6 PA, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0251 / 100.0	0.0296 / 96.7	0.0207 / 100.0
Organic fraction	0.0211 / 100.0	0.0296 / 96.7	0.0207 / 100.0

B.7 RX, P prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.0800 / 88.7	0.0791 / 73.3	0.0710 / 81.0
Conversion	0.0426 / 72.7	0.0321 / 70.0	0.0306 / 73.3
Safety risk	0.1031 / 58.3	0.0282 / 85.0	0.0370 / 81.7
Public score	0.1195 / 48.7	0.0501 / 83.3	0.0476 / 90.3
Selectivity	0.1198 / 58.0	0.0818 / 69.3	0.0714 / 77.3
Yield	0.1461 / 35.7	0.0708 / 76.7	0.0698 / 78.3

B.8 RX, P prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1509 / 57.7	0.0805 / 73.0	0.1271 / 60.3
Conversion	0.0474 / 85.3	0.0319 / 82.7	0.1960 / 53.0
Safety risk	0.0984 / 56.7	0.1106 / 56.7	0.0811 / 68.3
Public score	0.1341 / 30.0	0.0818 / 73.0	0.0879 / 68.3
Selectivity	0.1900 / 31.0	0.0923 / 69.7	0.1360 / 64.7
Yield	0.2128 / 19.3	0.1060 / 65.7	0.1499 / 68.7

B.9 RX, S prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1219 / 78.7	0.1212 / 75.3	0.1279 / 64.7
Conversion	0.0217 / 97.7	0.0266 / 95.3	0.0200 / 98.0
Safety risk	0.0664 / 80.0	0.0717 / 66.7	0.1276 / 43.3
Public score	0.0504 / 89.7	0.0816 / 61.3	0.0665 / 78.0
Selectivity	0.1280 / 72.7	0.1319 / 70.7	0.1400 / 55.0
Yield	0.1157 / 81.7	0.1196 / 75.7	0.1275 / 61.0

B.10 RX, S prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1376 / 63.7	0.1490 / 67.7	0.1188 / 66.3
Conversion	0.0370 / 88.7	0.0505 / 82.7	0.0425 / 88.3
Safety risk	0.1000 / 45.0	0.1255 / 68.3	0.1133 / 45.0
Public score	0.0932 / 56.0	0.0917 / 58.0	0.0637 / 81.0
Selectivity	0.1574 / 56.7	0.1570 / 60.3	0.1175 / 62.3
Yield	0.1485 / 63.3	0.1422 / 66.3	0.0975 / 73.0

B.11 EQ, P prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0129 / 96.3	0.0759 / 68.7	0.0646 / 74.7
Normalized pH	0.0031 / 94.3	0.0160 / 65.0	0.0150 / 68.7
Precipitation signal	0.0205 / 83.3	0.0376 / 73.3	0.0383 / 68.7

B.12 EQ, S prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0292 / 91.7	0.0235 / 76.3	0.0122 / 87.7
Normalized pH	0.0097 / 87.3	0.0200 / 75.7	0.0203 / 77.7
Precipitation signal	0.0033 / 95.3	0.0042 / 95.7	0.0078 / 87.3

B.13 EQ, E prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0128 / 83.7	0.0107 / 85.0	0.0134 / 79.0
Normalized pH	0.0233 / 80.7	0.0278 / 79.0	0.0202 / 82.0
Precipitation signal	0.0084 / 89.3	0.0116 / 92.0	0.0103 / 88.3

B.14 C, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.3279 / 25.0	0.3096 / 38.3	0.2925 / 43.3
Crystal purity	0.0095 / 98.3	0.0497 / 60.0	0.0917 / 23.3
Particle-size index	0.0777 / 55.0	0.0624 / 58.3	0.0725 / 58.3
Net crystal recovery	0.1632 / 48.3	0.0606 / 83.3	0.0892 / 71.7

B.15 C, E prior: delivery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.2281 / 51.7	0.2774 / 38.3	0.3046 / 41.7
Crystal purity	0.0197 / 95.0	0.0398 / 73.3	0.0508 / 55.0
Particle-size index	0.0566 / 71.7	0.0440 / 70.0	0.1023 / 48.3
Net crystal recovery	0.0911 / 61.7	0.0684 / 80.0	0.0786 / 73.3

B.16 P, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Purity	0.2240 / 51.7	0.1883 / 53.3	0.2508 / 50.0
Recovery	0.0668 / 56.7	0.0687 / 51.7	0.0727 / 53.3

C Exploratory analyses of response regimes

These analyses use the retained source campaigns and their original predictions. They do not define new experimental arms or alter the withheld-condition bank. The concentration grouping, joint crystallization contrast and illustrative traces were selected after the experiments, and are descriptive analyses of reused worlds.

C.1 Equilibrium: a reversal across public concentration regimes

The three most dilute recipes have nominal concentrations approximately 1.33×10^{-4} , 1.33×10^{-5} and 1.67×10^{-4} M. Each campaign has three normalized responses for each recipe. Point errors use five-observation means; interval coverage uses the individual reference observations. Group means average the corresponding responses and queries and then the five world campaigns. Weighting the two groups 3:9 reconstructs every original twelve-question campaign macro MAE. The other nine queries include boundary conditions; this split is not a predeclared interpolation-versus-extrapolation test.

Query group / arm	Macro MAE	Coverage (%)	Interval width	Source-mean MAE
Other nine / Opaque	0.01008	90.7	0.04706	0.00553
Other nine / Aligned	0.00465	91.3	0.03340	0.00518
Other nine / MisIndexed	0.00439	89.6	0.02798	0.00494
Three most dilute / Opaque	0.01833	93.3	0.15676	0.16961
Three most dilute / Aligned	0.15862	2.2	0.08084	0.16833
Three most dilute / MisIndexed	0.14397	13.8	0.09633	0.16836

Table C1. EQ/P subgroup results. Each row averages five worlds; rows reuse those worlds. The source-mean predictor is an additional post hoc diagnostic and obtains no new experiments. Its poor dilute-query performance shows why retaining the plateau is insufficient; its usefulness on the other nine depends on their response distribution.

Aligned improves on the other nine and worsens on the dilute three relative to Opaque in all five worlds. All fifteen source campaigns remain above 0.018 M. In the selected second-world comparison, Aligned spans about ninety-fold source concentration variation and Opaque about fourfold, but both remain in the observed plateau. At the most dilute question, reference dissociation is 0.65463; Opaque predicts 0.629 [0.45, 0.80] and Aligned 0.0804 [0.053, 0.108]. The Opaque prediction response develops an unsaturated weak-acid extrapolation beyond its sealed report. This reasoning must not be credited retrospectively to K1 or to an earlier source-stage hypothesis.

C.2 Crystallization: joint response changes under the same treatment

Budget / arm	Recovery MAE	Purity MAE	Recovery improves and purity worsens vs Opaque
12 / Opaque	0.16322	0.00952	Reference
12 / Aligned	0.06055	0.04969	5 / 5 worlds
12 / MisIndexed	0.08923	0.09166	5 / 5 worlds
24 / Opaque	0.09107	0.01975	Reference
24 / Aligned	0.06839	0.03983	2 / 5 worlds
24 / MisIndexed	0.07863	0.05080	3 / 5 worlds

Table C2. Joint directional comparisons use both response errors from the same world, arm and budget. At twelve batches, each joint direction also holds in all four pairs whose two campaigns conform to the source-assay contract. The direction weakens at 24 batches. A treatment can therefore be helpful on one response and harmful on another without implying a budget-invariant tradeoff.

C.3 Evidence status and interpretation

The same-source C references use only each campaign’s final assays and public recipe features. They exclude hidden truth during fitting but do not reproduce the agent’s complete intermediate measurements, prior or computational policy. Low held-out purity variation makes the mean a particularly strong reference. Neither its success nor the agent’s recovery advantage is a complete mechanism identification result.

Selected EC and C traces support detailed examples rather than population frequencies; the bounded EQ review below covers all fifteen campaigns in that block. The current evidence does not causally identify prior anchoring, report compression, acquisition-versus-inference mediation or a general preference for familiar chemistry. Full-history Q also prevents interpreting K1 as the sole information channel. Experimental selection and later interpretation are both part of the original autonomous process. Their separate causal contributions are not required to describe the observed whole-process contrasts and have not been identified here.

C.4 Original equilibrium campaigns: predictions and research chronology

We reviewed all fifteen retained parameter-prior equilibrium campaigns, comprising 180 batches, 945 operations and 45 K1/Q/K2 stages. Each row below is an original autonomous research session, not an independent reader of its records. The selected question uses 1 micromole in 75 mL; the same question and original reference targets apply to all three arms in a world. The reference column is the five-observation mean.

World / arm	Public Q explanation	Dissociation [80% interval]	Reference
1 / Opaque	Piecewise weak acid	0.6990 [0.250, 0.900]	0.7160
1 / Aligned	Plateau / weak trend	0.1040 [0.044, 0.225]	0.7160
1 / MisIndexed	Plateau / weak trend	0.1092 [0.0772, 0.1412]	0.7160
2 / Opaque	Piecewise weak acid	0.6290 [0.450, 0.800]	0.6546
2 / Aligned	Plateau / weak trend	0.0804 [0.053, 0.108]	0.6546
2 / MisIndexed	Plateau / weak trend	0.0806 [0.0556, 0.1056]	0.6546
3 / Opaque	Piecewise weak acid	0.5630 [0.430, 0.700]	0.5797
3 / Aligned	Plateau / weak trend	0.1100 [0.045, 0.190]	0.5797
3 / MisIndexed	Mixed weak acid / plateau	0.4600 [0.100, 0.750]	0.5797
4 / Opaque	Qualitative departure	0.5200 [0.150, 0.880]	0.5173
4 / Aligned	Plateau / weak trend	0.0724 [0.040, 0.105]	0.5173
4 / MisIndexed	Plateau / weak trend	0.0962 [0.040, 0.450]	0.5173
5 / Opaque	Piecewise weak acid	0.4130 [0.200, 0.700]	0.4480
5 / Aligned	Plateau / weak trend	0.0790 [0.035, 0.130]	0.4480
5 / MisIndexed	Plateau / weak trend	0.0681 [0.040, 0.096]	0.4480

Table C3. Public extrapolation accounts and original same-context predictions. Categories are a single-author retrospective reading of public explanations, not internal reasoning traces or independent causal labels. Four Opaque accounts use piecewise weak-acid reasoning; the fifth permits a qualitative departure. All five Aligned accounts extend a plateau or weak trend. The third-world MisIndexed mixed account is a counterexample to treating all dossier conditions alike.

The first-world Opaque K1 already estimates an effective acid constant near 2.2×10^{-5} and an active-pool cap near 0.004 M, while retaining competing interpretations. Its Q applies that hypothesis at low concentration. In the second world, Opaque K1 primarily supports the observed plateau and limits extrapolation; the unsaturated weak-acid calculation is first explicit in Q. The aligned K1 similarly limits its account to the sampled region, but its Q interval still misses the dilute response by a large margin. These differences separate what was stated before prediction from what became explicit during it.

All fifteen source concentration ranges lie above the three dilute test recipes. The exported per-operation decision-reason fields are unavailable for all 945 operations. Actions and observations can therefore be reconstructed, but a contemporaneous sequence of belief revisions cannot be inferred from these fields. K1, Q and K2 remain distinct public records rather than substitutes for that sequence.

In K2, eleven of fifteen campaigns propose a trace-load experiment: five Opaque, three Aligned and three MisIndexed. The other choices are equal-concentration scale-up, dilution near the supplied relation, comparison of instruments at the same state and repetition of an anomalous batch. K2 follows exposure to Q conditions without feedback on their reference outcomes. None of these proposals was executed; they are retrospective choices, not source-stage plans or demonstrated repairs.

D Selected autonomous research histories

Figure 2 and Supplementary Fig. S4 follow the same retrospectively selected Aligned crystallization pair. Each budget is an independent session. The case documents actions, observations and later public accounts; it does not estimate the cohort-wide effect of additional resources.

D.1 Experimental paths, recommendations and retests

A retrospectively selected Aligned crystallization pair illustrates how independent sessions can follow different experimental paths (Fig. 2). The 12-batch session first obtains a quality-feasible product in batch 1 and later lowers the cooling endpoint from 260 to 250 K. The 24-batch session first becomes feasible in batch 20, after changing heating and introducing staged cooling. Its selected procedure has higher independently retested recovery and lower fines than the smaller session’s selection. This example illustrates recorded actions and outcomes, not a population budget effect. Contemporaneous decision reasons were not recorded; the illustration therefore shows recorded changes without reconstructing immediate thoughts.

The 12-batch session first meets the quality constraints at batch 1, with recovery 34.7%, and reaches its best feasible observed recovery of 51.2% at batch 10. The 24-batch session first becomes feasible at batch 20 and reaches 57.0% at batch 23. Quality feasibility and subsequent optimization are distinct milestones. All 36 original batch results remain in the source data, including the infeasible trials.

Quench in this environment stops reaction chemistry while subsequent cooling and crystal growth remain possible. The two sessions differ in materials and multiple process conditions; their comparison does not isolate a quench effect. Heating and cooling values are requested targets, not measured temperature trajectories. The K1 summaries were written after the source experiments and are distinct from contemporaneous decision logs.

D.2 Sealed predictions and subsequent reflection

Figure S4 follows the same sessions through sealed prediction and subsequent reflection; the proposed experiment was not executed.

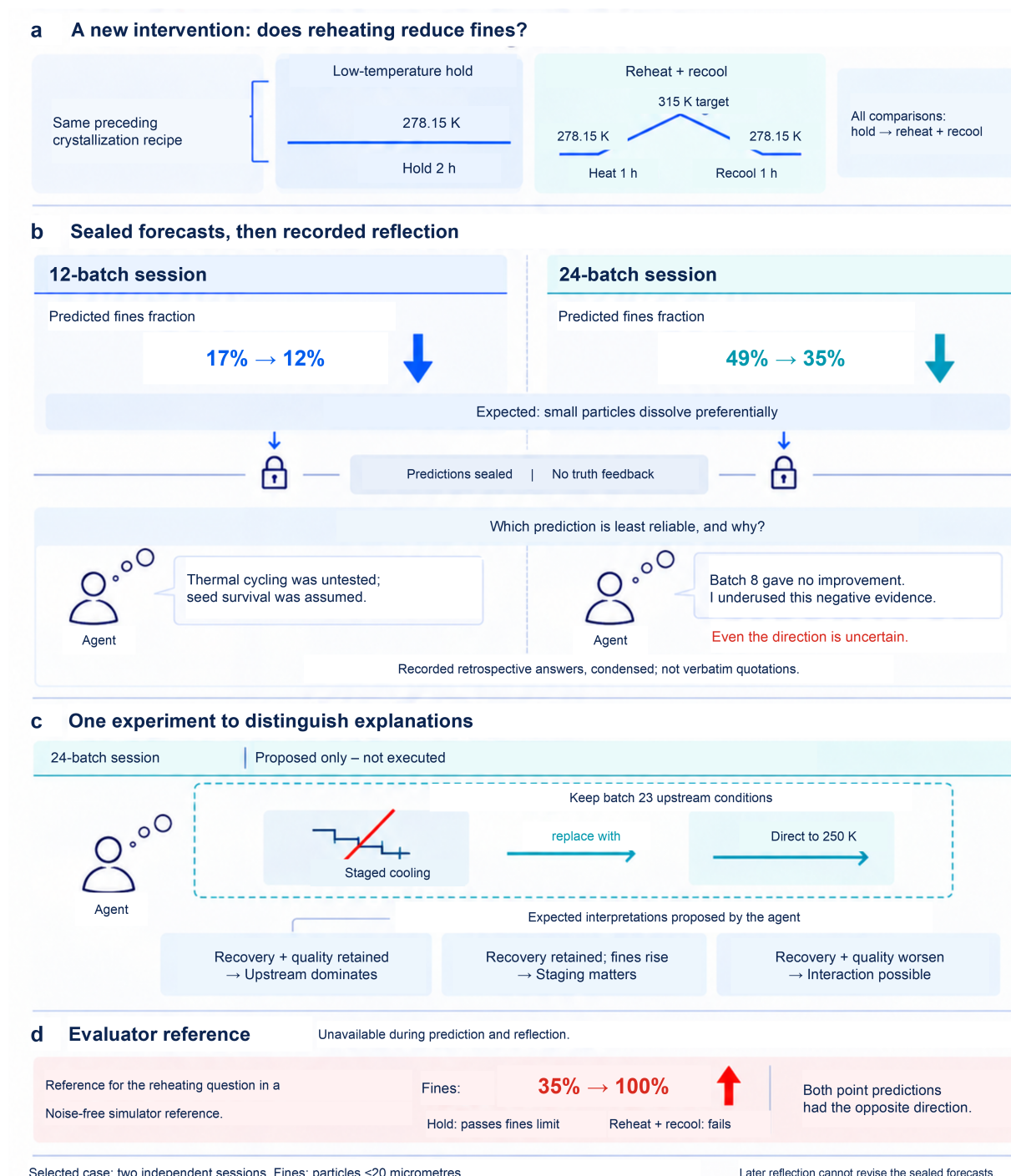


Figure S4: Sealed predictions and subsequent reflection in the same selected crystallization pair. a, The withheld comparison replaces a two-hour hold at 278.15 K with heating towards 315 K for one hour and recooling for one hour after a common preceding recipe. Thermal profiles are schematic. b, Both original sessions predict lower fines. The condensed public K2 answers follow sealed Q predictions without reference feedback. c, The 24-batch session proposes changing only the cooling procedure of its selected batch; possible outcomes would distinguish upstream, cooling and interaction explanations. This proposal was not executed. d, The evaluator's reference concerns the thermal intervention in a, not the proposed experiment in c, and was unavailable during Q and K2.

E Supporting outcomes and examples

E.1 Quality constraints and response-specific references

Purification gives a related distinction between recovery and acceptable delivery. Aligned recommendations meet the purity threshold in three of five worlds, compared with none for Opaque or MisIndexed. Yet mean recovered fractions are 0.04455, 0.12373 and 0.32105 for Aligned, Opaque and MisIndexed, respectively. Recovering more material therefore does not imply satisfying the commission. A single retest near the purity threshold is also not a precise estimate of a procedure's reliability. These system-specific outcomes cannot be represented faithfully by a universal yield score (Fig. S5a).

Purification provides another response-specific example: fourteen of fifteen campaigns correctly judge a small concentration effect on purity, but only two correctly judge the small purity effect of splitting a wash. Recognizing approximate invariance is itself a predictive achievement, and depends on the intervention.

In crystallization, agent size forecasts have lower MAE than the source mean in four of thirty campaigns and fines forecasts in twenty-one (Fig. S5b,c). Together with the recovery and purity contrasts in the main text, these results retain the complete response-specific comparison.

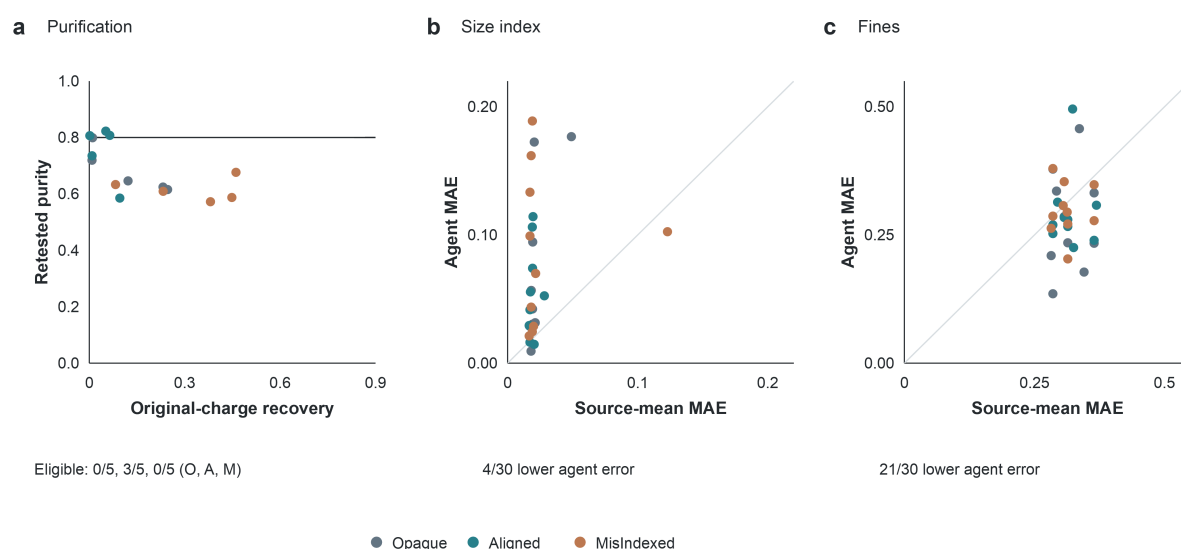


Figure S5: Additional operational and predictive outcomes. a, All fifteen purification recommendations, showing original-charge recovery and independently retested purity. The line marks the purity threshold of 0.80; eligible counts are 0/5, 3/5 and 0/5 for Opaque, Aligned and MisIndexed. b,c, Size and fines MAE for all thirty crystallization campaigns relative to each campaign's public observation mean. Points above the diagonal favour the mean; the campaign with a source-assay shortfall remains included. These panels use different response scales.

E.2 Additional selected examples

In a selected Aligned phase-partitioning campaign pair, the larger budget covers all sixteen material pairings and then allocates eight experiments to process variation; the smaller campaign covers ten pairings. This is a plausible route to improvement, not a cohort-wide causal explanation. Two Aligned world pairs nevertheless worsen.

Uncertainty is part of the same applicability problem. The poor dilute-equilibrium coverage shows that a wrong extrapolation can also be overconfident. In a retrospectively selected MisIndexed crystallization campaign, the mechanism report explicitly acknowledges that only one solvent was tested. Nevertheless, predicted purity averages 0.86833 against a reference mean of 0.98566, and all twelve nominal 80% intervals miss. A verbal statement of uncertainty does not by itself ensure calibrated quantitative prediction.

E.3 Complete crystallization baseline and interval comparisons

Figure 6b reports how often each response is predicted more accurately than the public observation mean. Table E1 supplies the corresponding error magnitudes and the public nearest-neighbour comparison. Each baseline is fitted separately to the same campaign’s public final assays; neither receives withheld outcomes or additional experiments. Nearest-neighbour prediction also uses public recipe features. These are empirical references rather than strong system-identification baselines.

Table E1. Crystallization prediction errors and baseline comparisons.

Response	Agent MAE	Mean MAE	Nearest MAE	Wins: mean	Wins: nearest
Recovery	0.09185	0.17135	0.17308	26/30	26/30
Purity	0.04354	0.00482	0.00905	0/30	4/30
Size index	0.06923	0.02345	0.02708	4/30	7/30
Fines fraction	0.29000	0.31651	0.32426	21/30	20/30

MAE is averaged equally over all thirty campaigns in each response’s original scale; errors are not ranked across responses. Recovery, purity and fines are fractions, while size is the declared size index. Mean and Nearest denote the public observation-mean and nearest-neighbour predictors. Wins count strictly lower agent MAE; neither comparison has ties. The source-assay shortfall remains included, and the five world instances are reused across arms and independent budget sessions.

Table E2 accompanies the purity coverage comparison in Figure 6d. Wider intervals need not cover a biased prediction’s reference, while coverage exceeding the nominal level does not by itself establish calibration or sharpness.

Table E2. Purity interval coverage and width by information arm and budget.

Budget	Information arm	Campaigns	Coverage (%)	Mean width (pp)
12	Opaque	5	98.3	6.81
12	Aligned	5	60.0	12.94
12	MisIndexed	5	23.3	15.53
24	Opaque	5	95.0	8.59
24	Aligned	5	73.3	12.37
24	MisIndexed	5	55.0	12.31

The original intervals have nominal 80% coverage and concern the pre-final-assay response target. Coverage and interval width are averaged over each campaign’s twelve queries and then over its five-world arm-budget group. Each row contains sixty query targets but only five world clusters; queries are not independent world replicates. Width is expressed in percentage points of purity. All original intervals and source failures remain unchanged.

F Complete most-dilute equilibrium predictions

F.1 Original intervals and campaign-specific observations

The main-text equilibrium comparison shows point predictions for the most dilute recipe. Table F1 retains all fifteen original 80% prediction intervals and each campaign’s source-response range. These are prediction intervals issued by the agent, not confidence intervals on an estimated mean. Each source range covers twelve final assays; the reference is the mean of the original five evaluator observations and was unavailable during prediction or retrospective reflection.

Table F1. Dissociation predictions at the most dilute recipe, with all values in percent.

World / arm	Source range	Point prediction	80% interval	Reference mean
1 / Opaque	6.18-8.34	69.90	[25.00, 90.00]	71.60
1 / Aligned	6.20-9.23	10.40	[4.40, 22.50]	71.60
1 / MisIndexed	6.11-9.25	10.92	[7.72, 14.12]	71.60
2 / Opaque	6.17-8.57	62.90	[45.00, 80.00]	65.46
2 / Aligned	6.12-8.06	8.04	[5.30, 10.80]	65.46
2 / MisIndexed	6.23-7.98	8.06	[5.56, 10.56]	65.46
3 / Opaque	5.94-8.22	56.30	[43.00, 70.00]	57.97
3 / Aligned	6.34-8.02	11.00	[4.50, 19.00]	57.97
3 / MisIndexed	6.16-7.94	46.00	[10.00, 75.00]	57.97
4 / Opaque	6.19-7.91	52.00	[15.00, 88.00]	51.73
4 / Aligned	5.82-8.14	7.24	[4.00, 10.50]	51.73
4 / MisIndexed	5.81-8.22	9.62	[4.00, 45.00]	51.73
5 / Opaque	5.49-7.48	41.30	[20.00, 70.00]	44.80
5 / Aligned	6.44-7.99	7.90	[3.50, 13.00]	44.80
5 / MisIndexed	6.58-7.79	6.81	[4.00, 9.60]	44.80

The recipe contains 1 micromole in 75 millilitres (nominal concentration 13.3 micromolar). The pooled research-observation band in the main figure is the minimum and maximum across all 180 assays; the ranges above are specific to each campaign. World 3 MisIndexed retains its mixed-explanation departure from the observed plateau. Public explanation categories and K1/Q chronology remain in section C.4.